MILITARY STANDARD

MIL-STD-1320-222
(NAVY)

30 OCTOBER 1985

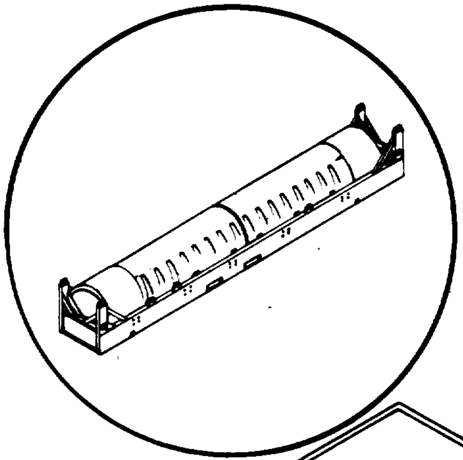
TRUCKLOADING
GUIDED MISSILE BGM-109
(TOMAHAWK-CLS)
IN SHIPPING SKID MK 30

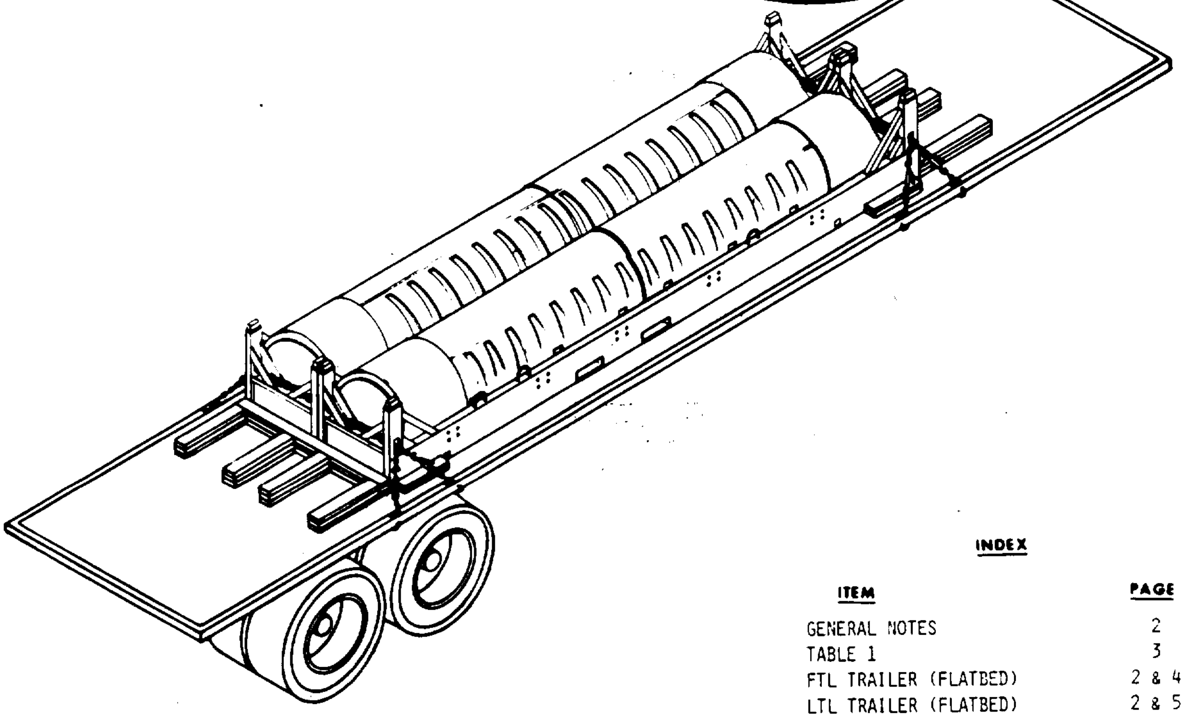
CONTAINER DATA

DIMENSIONS:

LENGTH - 307 5/8
WIDTH - 39 3/4
HEIGHT - 44
WEIGHT - SEE FTL NOTE 1
CUBE - 311.4 CU/FT

DOT HAZARD CLASSIFICATION - SEE TABLE 1





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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236(NAVY).

DISTRIBUTION STATEMENT A
Approved for public release; distribution is unlimited.

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY

W. D. Gray NWHC 10/30/85
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

12/4/85
SIGNATURE SEASYSCOM DIRECTION DATE

ORIGINATOR 30 Oct 85
DATE

NAVAL WEAPONS HANDLING CENTER
NWS EARLE, NEW JERSEY

GENERAL NOTES

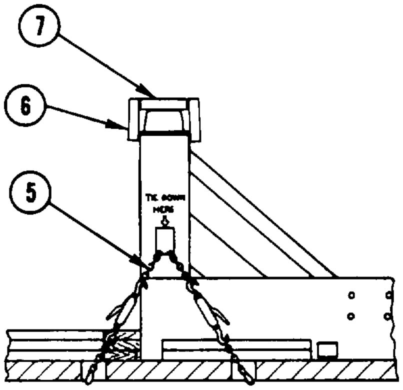
1. THIS DOCUMENT PROVIDES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE GUIDED MISSILE BGM-109 (TOMAHAWK-CLS) IN THE MK 30 SKID.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 35 FT AND LONGER FLATBED TRAILERS. THE TRAILERS SHALL HAVE THEIR AXLES LOCATED IN THE"WESTERN" POSITION (AT THE EXTREME REAR OF THE TRAILER).
3. CHAINS CAN BE USED FOR TIEDOWNS AND ARE THREADED ONLY THROUGH HOLES IN STACKING POSTS AS SHOWN IN DETAIL A.
4. CHAINS, FITTINGS AND LOAD BINDERS SHALL MEET THE REQUIREMENTS OF THE BASIC TRUCKLOADING DOCUMENT, MIL-STD-1320.
5. PRIOR TO LOADING THE TRAILER AND DURING PRELOADING INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS, THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH APPROPRIATE PLACARDS TO BOTH SIDES, FRONT AND BACK OF TRAILER (SEE OP 2165).
8. UNLESS OTHERWISE SPECIFIED, ALL MATERIAL SHALL BE AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320.

FTL & LTL (35 FT & LONGER TRAILERS)

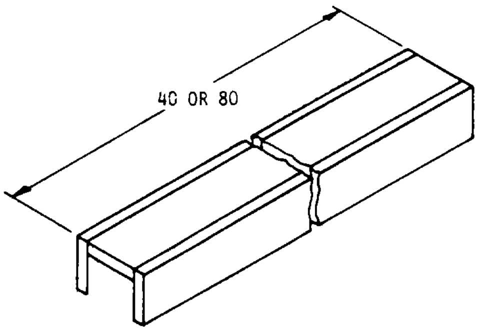
1. A FTL CONSISTS OF TWO CONTAINERS. TRUCKLOAD WEIGHT IS APPROXIMATELY 20,000 LBS. WEIGHT WILL VARY DEPENDING UPON CONFIGURATION OF MISSILE BEING LOADED.
2. A FTL OF TWO CONTAINERS SHALL HAVE FOUR TIE DOWNS AS SHOWN ON PAGE 3.
3. LOCATE CONTAINERS WITH EQUAL DISTANCE FROM ENDS OF TRAILER.
4. WHEN LOADING TWO CONTAINERS, PLACE SEPARATORS, ITEM 4, BETWEEN THEM AS SHOWN IN DETAIL B AND SEE WARNING BELOW
5. WHEN LOADING ONE CONTAINER, POSITION CONTAINER CENTERED ON TRAILER.
6. USE TWO TIEDOWNS WHEN LOADING ONE CONTAINER.
7. INSTALL TARPAULIN PROTECTOR BOARDS, SEE DETAIL C, AND TARPAULINS.

WARNING

WHEN LOADING THE SECOND OF TWO SKIDS ON THE TRAILER. IT IS ESSENTIAL THAT THE FORK LIFT TINES DO NOT EXTEND PAST THE FAR SIDE OF THE CONTAINER. THEREFORE ANY FORKLIFT EQUIPMENT MUST BE ABLE TO LIFT 9600 LBS WITH THE LOAD CG LOCATED 28 INCHES FROM THE FORKLIFT MAST. WHEN TRANSPORTING A SINGLE SKID, THE TINES CAN EXTEND PAST THE SKID BUT THE FORKLIFT MUST BE ABLE TO LIFT 9600 LBS WITH THE LOAD CG 48 INCHES FROM THE FORKLIFT MAST.



DETAIL Ⓐ



DETAIL Ⓒ

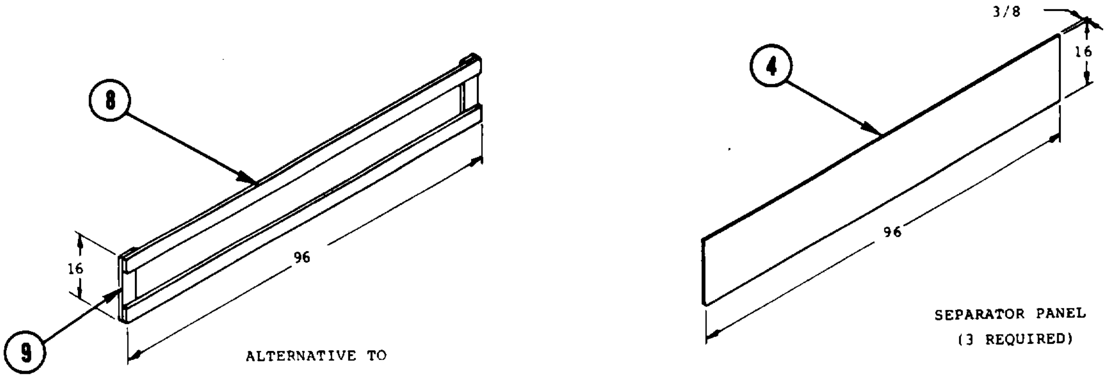
TABLE I

CONFIGURATIONS* OF TOMAHAWK
IN MK 45 CAPSULE
AUTHORIZED FOR SHIPMENT IN
MK 30 SKID

	MISSILE DESIGNATOR	BRIEF DESCRIPTION	DOT HAZARD CLASS	CG HAZARD CLASS
1.	BGM-109A-3 AND JBGM-109A-3	ENCAPSULATED TLAM		
	A.	W/O WARHEAD OR GAS GENERATOR	EXPL B	II-B
	B.	WITH WARHEAD	EXEMPT **	-
	C.	W/O WARHEAD, WITH OR WITHOUT REM	EXPL B	II-B
2.	BGM-109B-3 AND JBGM-109B-3	ENCAPSULATED TASM		
	A.	W/WARHEAD, AUR OR PRE-FLIGHT	EXPL A	II-A
	B.	W/O WARHEAD, WITH OR WITHOUT REM	EXPL B	II-B
3.	BGM-109C-3 AND JBGM-109C-3	ENCAPSULATED TLAM		
	A.	W/WARHEAD, AUR OR PRE-FLIGHT	EXPL A	II-A
	B.	W/O WARHEAD, WITH OR WITHOUT REM	EXPL B	II-B

AUTHORIZED CONFIGURATIONS INCLUDE ALL UP ROUNDS WITHOUT FUZING AND WITH VARIATIONS IN LIQUID FUEL LOADS.

**EXEMPT PER 49CFR173.7(3).



ALTERNATIVE TO
ITEM 4 (SEPARATOR PANEL)

SEPARATOR PANEL
(3 REQUIRED)

DETAIL B

TRUCKLOAD DATA

NUMBER OF CONTAINERS 2
WEIGHT .... (APPROX) - 20,000 LBS

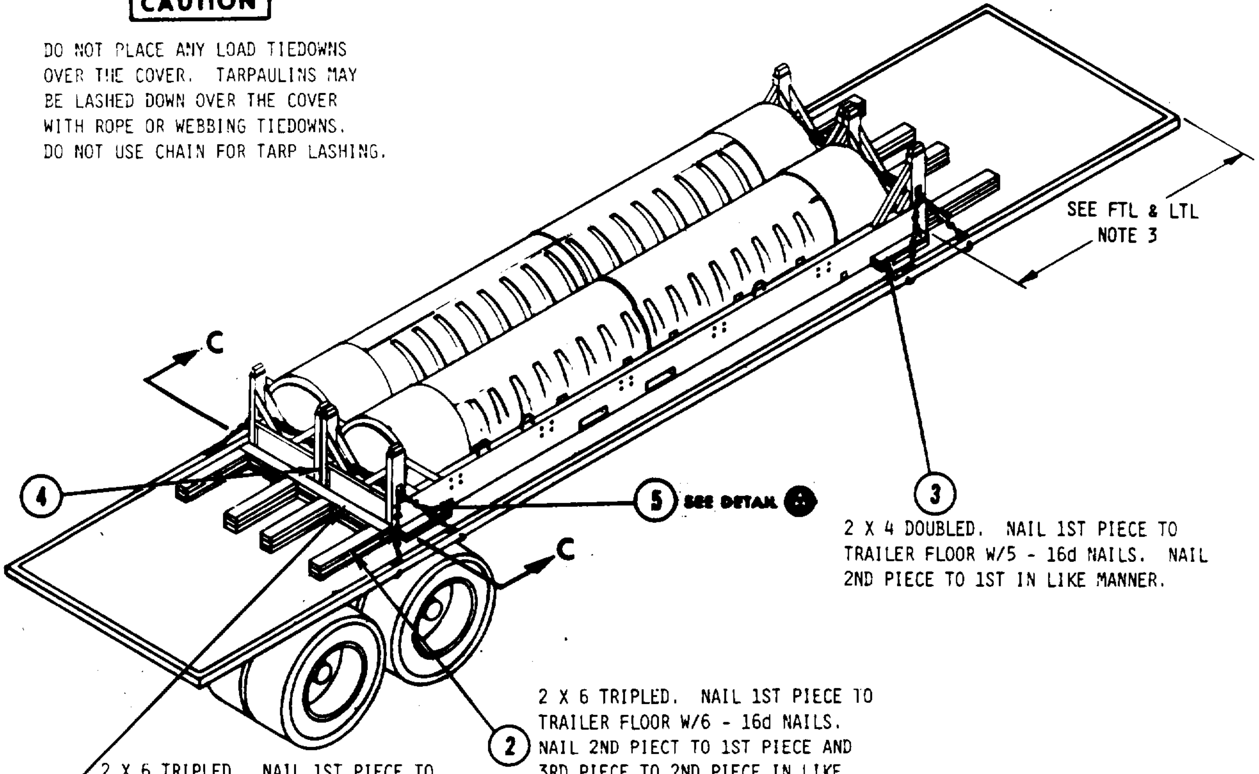
FTL NOTES:

(1) TARPAULINS AND PROTECTOR BOARDS OMITTED FOR CLARITY

(2) LOADBINDERS ARE SHOWN ON THE RIGHT SIDE OF TRAILER FOR CLARITY ONLY. PLACEMENT IS AT THE DISCRETION OF THE OPERATOR

CAUTION

DO NOT PLACE ANY LOAD TIEDOWNS OVER THE COVER. TARPAULINS MAY BE LASHED DOWN OVER THE COVER WITH ROPE OR WEBBING TIEDOWNS. DO NOT USE CHAIN FOR TARP LASHING.

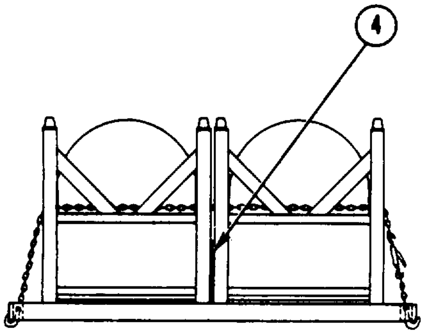


③ 2 X 4 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/5 - 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

② 2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/6 - 16d NAILS. NAIL 2ND PIECT TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER.

① 2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER

NOTE: 1. UNLESS OTHERWISE SPECIFIED, ALL MATERIALS SHALL BE IN ACCORDANCE WITH MIL-STD-1320.



SECTION C-C

6	9	VERTICAL SEPARATOR BOARD	SEE DETAIL B	1 X 4 X 16
6	8	HORIZONTAL SEPARATOR BOARD	SEE DETAIL B	1 X 4 X 96
2	7	HORIZONTAL TARP PROTECTOR	SEE DETAIL C	2 X 6 X 80
4	6	VERTICAL TARP PROTECTOR	SEE DETAIL C	2 X 6 X 80
4	5	CHAIN & BINDER	SEE NOTE 1	5/16 OR 3/8
3	4	SEPARATOR	PLYWOOD SEE NOTE 1	3/8 X 16 X 96
8	3	SLEEPER	SEE NOTE 1	2 X 4 X 24
24	2	BACKUP CLEAT	SEE NOTE 1	2 X 6 X 36
6	1	CROSSMEMBER	SEE NOTE 1	2 X 6 X 80
REQ'D	**ITEM**	**DESCRIPTION**	**MAT'L/DWG**	**DIMENSIONS**

LIST OF MATERIALS

FTL 40 FT TRAILER (FLATBED)

TRUCKLOAD DATA

NUMBER OF CONTAINERS - 1
WEIGHT (APPROX) - 10,000 LBS

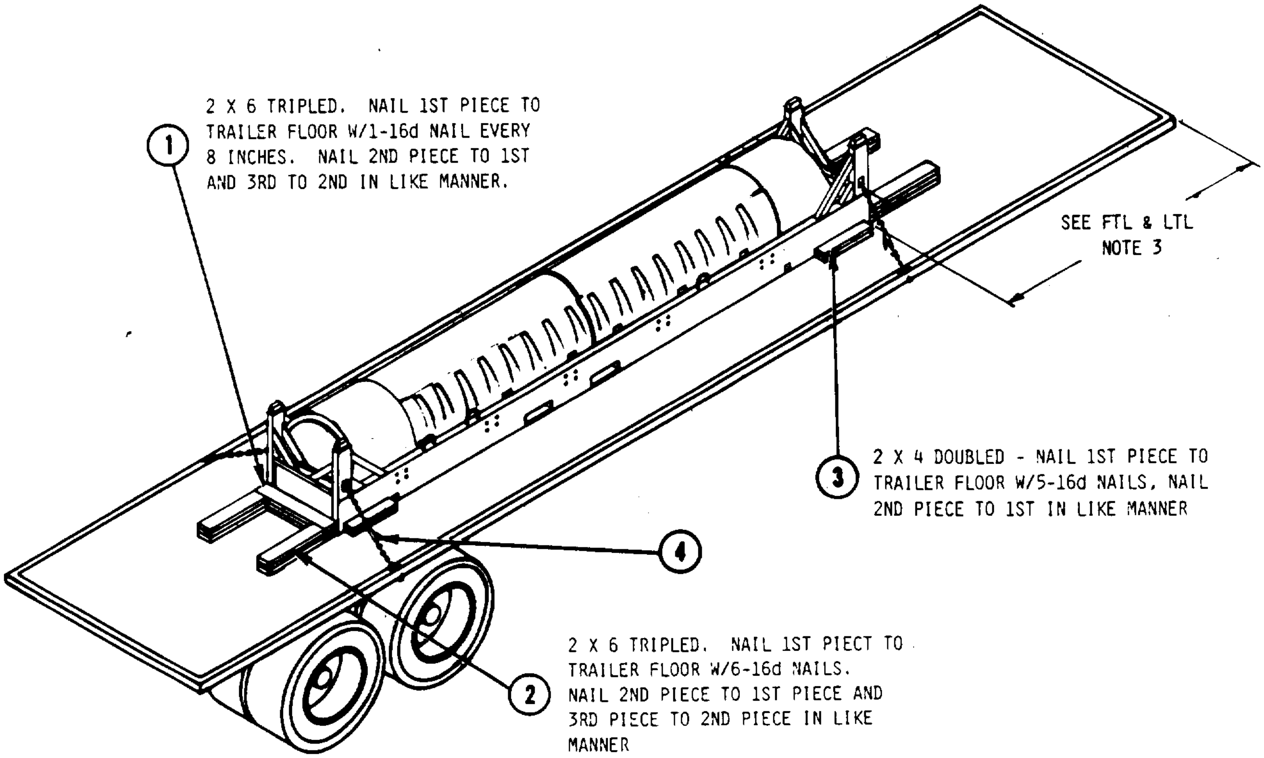
LTL NOTES:

(1) TARPAULINS AND PROTECTOR BOARDS OMITTED FOR CLARITY.

(2) LOADBINDERS ARE SHOWN ON THE RIGHT SIDE OF THE TRAILER FOR CLARITY ONLY. PLACEMENT IS AT THE DISCRETION OF THE OPERATOR.

CAUTION

DO NOT PLACE ANY LOAD TIEDOWNS OVER THE COVER. TARPAULINS MAY BE LASHED DOWN OVER THE COVER WITH ROPE OR WEBBING TIEDOWNS. DO NOT USE CHAIN FOR TARP LASHING.



① 2 X 6 TRIPLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST AND 3RD TO 2ND IN LIKE MANNER.

SEE FTL & LTL NOTE 3

③ 2 X 4 DOUBLED - NAIL 1ST PIECE TO TRAILER FLOOR W/5-16d NAILS, NAIL 2ND PIECE TO 1ST IN LIKE MANNER

② 2 X 6 TRIPLED. NAIL 1ST PIECT TO TRAILER FLOOR W/6-16d NAILS. NAIL 2ND PIECE TO 1ST PIECE AND 3RD PIECE TO 2ND PIECE IN LIKE MANNER

NOTE:
UNLESS OTHERWISE SPECIFIED, ALL MATERIALS SHALL BE IN ACCORDANCE WITH MIL-STD-1320.

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
2	6	HORIZONTAL TARP PROTECTOR	SEE DETAIL C	2 X 6 X 40
4	5	VERTICAL TARP PROTECTOR	SEE DETAIL C	2 X 6 X 40
2	4	CHAIN & BINDER	SEE NOTE 1	5/16 OR 3/8
8	3	SLEEPER	SEE NOTE 1	2 X 4 X 24
12	2	BACKUP CLEAT	SEE NOTE 1	2 X 6 X 36
6	1	CROSSMEMBER	SEE NOTE 1	2 X 6 X 40

LIST OF MATERIALS

LTL TRAILER (FLATBED)

REVIEW ACTIVITY:
NAVY-OS

PREPARING ACTIVITY:
NAVY-OS
(PROJECT NO. 8140-N652)